Derivatives Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. The four derivatives and their payoffs
2. How often interest is added
3. Forward price by no-arbitrage
4. Futures, margin and daily settlement
5. Hedging with futures
6. Forward rates, duration and convexity
7. Swaps: trading payment streams
8. Option strategies and put-call parity
9. Binomial trees for options
10. Black-Scholes and the Greeks

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: The four derivatives and their payoffs

A derivative is a contract whose value comes from something else (a stock, a rate, oil). There are four basic kinds: forwards, futures, swaps and options.

Start with an everyday idea

Imagine agreeing today to buy a bicycle next month for $100. A forward is that commitment. An option is different: you pay for the right to choose later whether to buy.

Words you need

Underlying means the thing the contract depends on. Long means you bought the contract; short means you sold it. A call is a right to buy; a put is a right to sell. Strike is the agreed option price.

Build the explanation

Forward: agree today to buy (long) or sell (short) at a fixed price K on a later date. Both sides must go through with it. Made privately (over the counter).

Future: the same idea, but traded on an exchange, standardized, and settled every day through margin (Class 7).

Swap: trade one stream of payments for another, e.g. fixed interest for floating (Class 9).

Option: the right but not the obligation. A call is the right to buy at K, a put the right to sell at K. You pay a premium up front for that right.

Payoff is what you get at the end. Profit is payoff minus what you paid. Draw the picture every time; it stops sign mistakes.

Formula and units

Long forward payoff = S_T - K
Short forward payoff = K - S_T
Long call payoff = max(S_T - K, 0)
Long put payoff = max(K - S_T, 0)
Profit = payoff - premium paid

S_T = price of the asset at the end, K = agreed (strike/delivery) price.

Walk through the sketch

1. Long forward: you win $1 for every $1 the price ends above K, and lose the same below.

2. Short forward: the mirror image. Your gain is exactly the other side's loss.

3. Long call: you only use it if the price is above K. Dashed line = profit after paying the premium.

4. Long put: you profit when the price falls below K. Insurance for a stock owner.

5. Four shapes to remember: two straight lines, two "hockey sticks".

Worked example

You buy a call with strike $100 for a $5 premium. At expiry the stock is $120. What is your payoff and profit? What if the stock is $90?

1. At $120: payoff = max(120 - 100, 0) = 20

2. Profit = 20 - 5 = 15

3. At $90: payoff = max(90 - 100, 0) = 0 (you just do not use the option)

4. Profit = 0 - 5 = -5. The most a call buyer can lose is the premium.

At $120: payoff 20, profit 15. At $90: payoff 0, profit -5

Common mistakes

1. Mixing payoff with profit.

2. Saying a forward costs money to enter. A fair forward costs nothing up front.

3. Forgetting the short side of an option can lose a lot.

Try before checking

You own a call with strike $100. The asset ends at $120. Its payoff is...

1. $20

2. $120

3. $0

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_02_Lecture_01_Introduction_to_Derivatives.pdf; Hull 8e, Chapters 1 and 10

Lesson 2: How often interest is added

"10% a year" can mean different amounts of money depending on how often interest is added. Continuous compounding is the limit where it is added every instant.

Start with an everyday idea

Interest can earn interest. At 10% a year, $100 becomes $110 after one year and $121 after two, because the second year also earns interest on the first $10.

Words you need

Compounding means adding interest to the balance. Annual means once per year. Continuous compounding is a mathematical limit, not a bank paying you every instant.

Helpful earlier topics: Money today vs money later

Build the explanation

If the bank adds interest twice a year, you earn interest on your interest sooner, so you end up with a bit more. Same headline rate, different result. That is why every rate needs a compounding frequency attached.

m times a year: grow by (1 + R/m) each piece, m pieces a year.

Continuous: grow by eRt. Derivatives math (forwards, Black-Scholes) almost always uses this.

Converting: two rates are "the same" if they grow $1 to the same amount in one year.

Formula and units

Compounded m times: FV = A × (1 + R/m)^(m·t)
Continuous: FV = A × e^(R·t)
Convert m → cont: Rc = m × ln(1 + Rm/m)
Convert cont → m: Rm = m × (e^(Rc/m) - 1)

A = amount, R = annual rate, m = times per year, t = years, ln = natural log.

Walk through the sketch

1. $100 at 10% a year. Added once a year, it jumps to $110 at the end.
2. Added twice a year: $105 after 6 months, then 5% on $105 gives $110.25.
3. Monthly: 12 tiny jumps. You end at $110.47.
4. Continuous: a smooth curve, interest every instant. You end at $110.52.

5. The more often you compound, the more you get, but it levels off at e^0.10.

Worked example

You have 10% a year compounded twice a year. What is the continuously compounded rate?

1. Each half year you grow by 1 + 0.10/2 = 1.05

2. Use the convert rule: Rc = 2 × ln(1.05)

3. ln(1.05) = 0.04879, times 2 = 0.09758

4. Check: e^0.09758 = 1.1025, and 1.05^2 = 1.1025. Same growth.

Rc = 9.758% continuous

Common mistakes

1. Plugging a semiannual rate into e^(rT) without converting it.

2. Using t in months when the rate is per year.

3. Thinking a higher m changes the headline rate. It only changes how much you actually earn.

Try before checking

At 10% annual compounding, $100 becomes what after two years?

1. $120

2. $121

3. $110

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx; Hull 8e, Chapter 4 (Interest Rates)

Lesson 3: Forward price by no-arbitrage

The fair forward price is what it costs to buy the asset now with borrowed money and carry it to the delivery date. Any other price gives someone free money.

Start with an everyday idea

If buying an item now and storing it costs $105 by next year, a promise to deliver the same item next year should not cost $120 when you can freely buy, finance and deliver it yourself.

Words you need

Spot is the price now. Forward price is the delivery price agreed today. Arbitrage is a locked-in gain from inconsistent prices under the model’s trading assumptions.

Helpful earlier topics: Money today vs money later, How often interest is added, The four derivatives and their payoffs

Build the explanation

Arbitrage means a riskless profit with no money down. Markets hunt it away. So: if you can borrow $40 at 5%, buy the stock, and hold it a year, it costs you $40 × e^0.05 = $42.05 to own the stock in a year. A forward to buy it in a year must also cost $42.05, otherwise you would do one and the opposite of the other.

Income (dividends) makes holding the stock cheaper, so subtract its PV: F = (S - I)erT.

Dividend yield q or a foreign interest rate rf works the same: F = S e(r-q)T.

Forward price vs forward value: a new forward is worth 0. An old one with delivery price K is worth (F0 - K)e-rT to the long side.

Formula and units

No income: F0 = S0 × e^(rT)
Known income I: F0 = (S0 - I) × e^(rT)
Yield q / FX: F0 = S0 × e^((r - q)T) (q = r_f for currency)
Value of long: f = (F0 - K) × e^(-rT)

S0 = spot price now, r = risk-free rate (continuous), T = years, K = delivery price written in an existing contract.

Walk through the sketch

1. Today the stock costs $40. You can borrow at 5%.
2. Strategy: borrow $40 and buy the stock. Costs you nothing today.
3. In one year you own the stock and owe 40 × e^0.05 = $42.05.

4. If someone offers a forward at $43, sell it to them. Deliver the stock, get 43, repay 42.05. Free $0.95.

5. So the only price with no free money is F0 = $42.05.

Worked example

A stock is $40, pays no dividend, r = 5%, T = 1 year. (a) Fair forward price? (b) The market quotes $43. How do you make money? (c) An old contract has K = $40. What is it worth?

1. (a) F0 = 40 × e^0.05 = 42.05

2. (b) $43 is too high. Sell the forward at 43, borrow $40, buy the stock.

3. In a year: deliver the stock, receive 43, repay the loan 42.05. Profit 0.95 with no risk.

4. (c) f = (42.05 - 40) × e^(-0.05) = 1.95 for the long side.

F0 = 42.05; arbitrage profit 0.95; old contract worth 1.95

Common mistakes

1. Mixing price (the K that makes value 0) and value (what an existing contract is worth).

2. Adding dividends instead of subtracting them.

3. Doing the arbitrage in the wrong direction. Forward too high means sell the forward and buy the asset.

Try before checking

With no income or storage cost, spot $100 and one-year simple interest 5%, the forward price is...

1. $95

2. $100

3. $105

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx; FRE6073_SOLUTION_03_Assignment_01_Solutions.pdf

Lesson 4: Futures, margin and daily settlement

With futures, gains and losses are paid every day through a margin account, so nobody can build up a big unpaid loss.

Start with an everyday idea

A futures exchange settles gains and losses each day. Even a sensible hedge can require cash before its final benefit arrives.

Words you need

Margin is collateral, not a down payment on the asset. Daily settlement transfers that day’s gain or loss. A margin call asks you to replenish collateral.

Helpful earlier topics: The four derivatives and their payoffs, Forward price by no-arbitrage

Build the explanation

Initial margin: the deposit to open the position.

Marking to market: each day the exchange moves money between buyers and sellers based on the new futures price.

Maintenance margin: a floor. If your account drops below it you get a margin call and must top it back up to the initial margin.

Clearing house: stands between buyer and seller, so you never depend on the other trader paying. That removes most counterparty risk.

Basis = spot price - futures price. It goes to zero at delivery.

Formula and units

Daily gain (long) = (F_today - F_yesterday) × contract size × number of contracts
If balance < maintenance → deposit (initial - balance)

Short positions get the opposite sign.

Walk through the sketch

1. You open with $12,000. The dashed red line is the $9,000 maintenance floor.
2. Day 1: gold falls $9. You lose $1,800 that day. Balance $10,200.
3. Day 2: down $3 more. Balance $9,600.
4. Day 3: down $5. Balance $8,600 is under the floor. Margin call! Add $3,400 to get back to $12,000.

5. Day 4: gold rises $3. You gain $600. Balance $12,600.

Worked example

You go long 2 gold futures (100 oz each) at $1,250. Initial margin $6,000 per contract, maintenance $4,500. Prices: 1,241 → 1,238 → 1,233 → 1,236. Track the account.

1. Start: 2 × 6,000 = $12,000. Each $1 move = 2 × 100 = $200.

2. Day 1: -9 × 200 = -1,800 → $10,200

3. Day 2: -3 × 200 = -600 → $9,600

4. Day 3: -5 × 200 = -1,000 → $8,600, below the $9,000 floor. Margin call: deposit $3,400 to get back to $12,000.

5. Day 4: +3 × 200 = +600 → $12,600

Margin call of $3,400 on day 3; balance $12,600 after day 4

Common mistakes

1. Topping up only to the maintenance level. You go back to the initial margin.

2. Forgetting to multiply by contract size and number of contracts.

3. Thinking margin is a fee. It is your money, held as a deposit.

Try before checking

You are long one futures contract with multiplier 10. Price falls from 100 to 98. Today’s change is…

1. $20 loss

2. $20 gain

3. $2 loss

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx; Hull 8e, Chapter 2

Lesson 5: Hedging with futures

A hedge uses futures to cancel out a price risk you already have. The hedge ratio tells you how many futures to use when the futures and your exposure do not move one-for-one.

Start with an everyday idea

A farmer wants to reduce uncertainty about the selling price of a crop. A futures position can offset some price movements, but the hedge may not match the crop perfectly.

Words you need

A hedge offsets a risk you already have. Basis is spot price minus futures price. Hedge ratio is the amount of protection relative to the exposure.

Helpful earlier topics: Futures, margin and daily settlement, Volatility, diversification, VaR and ES

Build the explanation

Short hedge: you will sell something later (a farmer's wheat). Sell futures now so a price fall is offset by a futures gain.

Long hedge: you will buy something later (an airline's fuel). Buy futures now.

Basis risk: basis = spot - futures. If you hedge with a different but related product (heating oil for jet fuel), or close out early, the basis can move and the hedge is not perfect.

Minimum-variance hedge ratio h*: the slope when you regress changes in your price on changes in the futures price. It is $\rho \times \sigma S/\sigma F$.

Number of contracts: h* × (size of your exposure) / (size of one contract).

Stock index hedge: to remove market risk from a portfolio with beta β, short β × (portfolio value)/(futures price × multiplier) contracts.

Formula and units

$h^* = \rho \times \sigma S / \sigma F$

$N^* = h^* \times QA / QF$

Index hedge: $N^* = \beta \times VA / VF$

σS, σF = std dev of price changes (spot, futures), ρ = their correlation, QA = units exposed, QF = units per contract, VA = portfolio value, VF = futures price × multiplier.

Walk through the sketch

1. Each dot is one month: how much jet fuel moved (up) vs how much heating oil futures moved (across).

2. They move together, but not exactly one-for-one.

3. The best-fit line has slope 0.78. That slope is the hedge ratio h*.

4. So hedge 78% of the 2m gallons: 1.56m gallons ÷ 42,000 = 37 contracts.

Worked example

An airline will buy 2 million gallons of jet fuel. It hedges with heating oil futures (42,000 gallons each). σS = 0.0263, σF = 0.0313, ρ = 0.928. How many contracts?

1. h* = 0.928 × 0.0263 / 0.0313 = 0.7798

2. Exposure to hedge = 0.7798 × 2,000,000 = 1,559,520 gallons

3. Contracts = 1,559,520 / 42,000 = 37.13

4. Round: buy 37 heating oil futures (a long hedge, since it will buy fuel).

h* ≈ 0.78, buy about 37 contracts

Common mistakes

1. Using h = 1 when the futures is on a different product.

2. Going short when you need to buy later (it is a long hedge).

3. Forgetting that a hedge also removes your upside.

Try before checking

You own an asset and fear a price fall. A basic futures hedge is…

1. Sell futures

2. Buy more of the same asset

3. Always buy futures

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx; Hull 8e, Chapter 3 (Hedging strategies using futures)

Lesson 6: Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

Start with an everyday idea

A fixed-rate bond is less attractive when new bonds pay more. Its price tends to fall. Duration estimates how strongly its price responds to a small rate change.

Words you need

Yield is a rate implied by price and payments. Modified duration measures price sensitivity. Convexity adjusts for curvature. A forward rate is a rate implied today for a future period.

Helpful earlier topics: Bonds and building a zero curve

Build the explanation

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always underestimates the price after a big move either way. Full repricing is the true answer; duration is the quick estimate.

Formula and units

Forward rate: $F(T1,T2) = (R2 \cdot T2 - R1 \cdot T1) / (T2 - T1)$

Duration: $D = \Sigma\ t \times [CF(t) \cdot e^{(-y \cdot t)}] / B$

Price change: $\Delta B \approx -B \times D \times \Delta y\ (+ \frac{1}{2} \times B \times C \times \Delta y^2)$

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Walk through the sketch

1. The price of the bond at each yield. It slopes down and bends (is "convex").
2. At today's yield of 12%, the price is $94.21.
3. Duration draws the tangent line at that point. Slope = -B × D.
4. The tangent is below the curve on both sides. That gap is convexity.
5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

Worked example

3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.

2. Discount each at e^(-0.12t) and add: B = 94.213

3. Weight each time by its share of the price and add: D = 2.653 years

4. ΔB ≈ -94.213 × 2.653 × 0.001 = -0.250, so the price falls to about 93.963.

B = 94.21, D = 2.653, price drops about $0.25

Common mistakes

1. Using Δy = 0.1 instead of 0.001.

2. Forgetting the minus sign: rates up means price down.

3. Using duration for a large rate shock without the convexity term.

Try before checking

Modified duration is 5 and yield rises 0.01 (one percentage point). Approximate price change?

1. -5%

2. +5%

3. -0.05%

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx; Risk syllabus RMN2 (duration, convexity, term structure)

Lesson 7: Swaps: trading payment streams

In an interest rate swap, one side pays fixed and receives floating. You value it as owning one bond and owing another.

Start with an everyday idea

Two borrowers can agree to exchange interest payments: one pays a fixed rate, the other a changing rate. They usually exchange only the net difference in a same-currency interest-rate swap.

Words you need

Notional is the reference amount used to calculate payments. Fixed stays agreed; floating resets to a market rate. A payment stream is a series of payments over time.

Helpful earlier topics: Money today vs money later, Forward rates, duration and convexity

Build the explanation

Company A has a floating-rate loan but wants a fixed payment. It enters a swap: pays a fixed 3% to the bank, receives the floating rate back. Only interest moves; the notional (e.g. $100m) never changes hands in an interest rate swap.

Bond view: receive floating + pay fixed = long a floating bond, short a fixed bond. Value = Bfloat - Bfixed.

Right after a reset, a floating bond is worth its par value (100).

Forward view: treat each future exchange as a forward rate agreement and add them up. Both views give the same answer.

Currency swap: swap interest and principal in two currencies. Value each leg in its own currency, convert at today's spot rate.

Formula and units

Pay fixed, receive floating: $V = B_{float} - B_{fixed}$

$B_{fixed} = \Sigma\, c \cdot e^{(-r_i \cdot t_i)} + L \cdot e^{(-r_n \cdot t_n)}$

$B_{float} = L$ (just after a reset)

L = notional, c = fixed coupon payment, r_i = zero rate for time t_i.

Walk through the sketch

1. Two parties: Company A and a bank.

2. A pays the bank a fixed 3% of $100m every year: $3m.

3. The bank pays A the floating rate (e.g. SOFR) on $100m.

4. Same as A owning a floating bond (worth 100) and owing a fixed bond (worth 99.93).

5. Swap value to A = 100 - 99.93 = +0.07 ($m).

Worked example

$100m swap, 2 years left, you pay 3% fixed annually and receive floating (just reset). Zero rates: 1-yr 2.5%, 2-yr 3.0% (continuous). Value to you?

1. Fixed bond: 3 × e^(-0.025) + 103 × e^(-0.06)

2. = 2.926 + 97.001 = 99.928 ($m)

3. Floating bond right after reset = 100

4. Value = 100 - 99.928 = +0.072 ($m). Positive: the 3% you pay is a little below the fair swap rate for this curve (about 3.04%), so the swap is slightly in your favour.

Swap worth about +$72,000 to the fixed payer

Common mistakes

1. Getting the sign backwards. Write which side you are on first.

2. Forgetting the notional at the end of the fixed bond (103, not 3).

3. For currency swaps, converting with forward rates in the bond view. Use spot.

Try before checking

On $1,000 for one year, you pay 5% fixed and receive 6% floating. Net cash flow?

1. $10 received

2. $10 paid

3. $60 received

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx; FRE6073_ASSIGNMENT_05_Assignment_03.pdf

Lesson 8: Option strategies and put-call parity

Options can be stacked like Lego to make any payoff shape you want. Put-call parity is the rule that links a call and a put with the same strike.

Start with an everyday idea

Combining options is like combining pieces of a safety net: one part limits losses, another can cap gains. Start by adding the payoff of every piece at each possible final price.

Words you need

Premium is the option’s purchase price. A spread combines options. Put-call parity relates matching European calls, puts, stock and a risk-free payment under stated assumptions.

Helpful earlier topics: The four derivatives and their payoffs, Forward price by no-arbitrage

Build the explanation

Bull call spread: buy a low-strike call, sell a high-strike call. Cheaper than a call alone, but profit is capped. You win if the price rises.

Bear spread: the opposite, for a falling price.

Straddle: buy a call and a put at the same strike. You win if the price moves a lot either way. A bet on volatility.

Strangle: like a straddle with two different strikes. Cheaper, needs a bigger move.

Butterfly: buy calls at K1 and K3, sell two at K2. Wins if the price stays near K2 (Class "Volatility smiles" uses it to find probabilities).

Covered call / protective put: stock + short call, or stock + long put (insurance).

Put-call parity: a call plus cash K·e-rT pays the same as a put plus the stock, so they must cost the same. Only for European options.

Bounds: a call is worth at least S - Ke-rT and at most S.

Formula and units

Put-call parity: $c + K \cdot e^{(-rT)} = p + S0$

Call bounds: $\max(S0 - K \cdot e^{(-rT)}, 0) \le c \le S0$

Put bounds: $\max(K \cdot e^{(-rT)} - S0, 0) \le p \le K \cdot e^{(-rT)}$

Profit of a strategy = sum of the payoffs - net premium paid.

Walk through the sketch

1. Start: buy a $30 call. Hockey stick that starts at 30.

2. Add: sell a $35 call. That flips the slope back to flat above 35.

3. Together: the bull spread. Profit is capped at 3, loss is capped at 2.

4. Different idea: a straddle (call + put at the same strike) is a V. It wins on big moves either way.

Worked example

(a) Buy a $30 call for $3 and sell a $35 call for $1. Max profit and max loss? (b) A European call with K = 30, T = 3 months costs $3. S = $31, r = 10%. What must the put cost?

1. (a) Net cost = 3 - 1 = 2. Max loss = 2 (price ends below 30).

2. Max payoff = 35 - 30 = 5, so max profit = 5 - 2 = 3 (price ends above 35).

3. (b) p = c + K·e^(-rT) - S = 3 + 30·e^(-0.025) - 31

4. = 3 + 29.259 - 31 = 1.26

Spread: max profit 3, max loss 2. Put = $1.26

Common mistakes

1. Using put-call parity for American options (it only gives bounds there).

2. Forgetting to subtract the premium when drawing profit.

3. Mixing up which leg is bought and which is sold in a spread.

Try before checking

An option payoff is $12 and you paid $5. Profit, ignoring financing?

1. $12

2. $7

3. $17

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Hull 8e, Chapters 10-11 (Properties of options, trading strategies); FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf

Lesson 9: Binomial trees for options

Let the stock only go up or down each step. Then you can copy the option exactly with stock and cash, and that copy's cost is the option price.

Start with an everyday idea

Replace a complicated future with two possible next prices: up or down. Calculate the option payoff in each state, then work backward to find today’s model price.

Words you need

A node is a point in the tree. Replication means matching cash flows using other assets. Risk-neutral probability is a pricing weight, not a prediction of real-world odds.

Helpful earlier topics: Money today vs money later, The four derivatives and their payoffs

Build the explanation

Replication: hold Δ shares and borrow some cash so the portfolio pays the same as the option in both the up and down state. Δ = (fu - fd)/(Su - Sd).

Risk-neutral probability p: the made-up probability that makes the stock earn the risk-free rate. It is a pricing tool, not anyone's real forecast.

Backward induction: start at the end with payoffs, step back one node at a time: value = e-rΔt[p·fu + (1-p)·fd].

American options: at each node, value = max(hold value, exercise now). Matters most for puts.

With many small steps the tree converges to Black-Scholes (Class 17). Typical choice: u = eσ√Δt, d = 1/u.

Formula and units

p = (e^(rΔt) - d) / (u - d)
f = e^(-rΔt) × [p·f_u + (1 - p)·f_d]
Δ = (f_u - f_d) / (S·u - S·d)
American: f = max(f, exercise value)

u, d = up and down factors, Δt = length of one step, f = option value at a node.

Walk through the sketch

1. Today the stock is $20.

2. After 3 months it is either $22 (up 10%) or $18 (down 10%).

3. After 6 months: $24.20, $19.80 or $16.20.

4. At the end, a call with strike $21 pays 3.20, 0, 0.

5. Step back using p = 0.6523 and discount: 2.03 at the up node, then $1.28 today.

Worked example

Stock $20, u = 1.1, d = 0.9, r = 12%, each step is 3 months. Price a European call with strike $21: one step, then two steps.

1. p = (e^(0.12×0.25) - 0.9) / 0.2 = (1.0305 - 0.9)/0.2 = 0.6523

2. One step: up price 22 → payoff 1; down 18 → payoff 0. c = e^(-0.03) × 0.6523 × 1 = 0.633 (Δ = 1/4 = 0.25 shares)

3. Two steps: ends 24.2, 19.8, 16.2 → payoffs 3.2, 0, 0

4. Up node: e^(-0.03) × 0.6523 × 3.2 = 2.0257; down node = 0

5. Today: e^(-0.03) × 0.6523 × 2.0257 = 1.2822

One-step call = $0.633; two-step call = $1.2822

Common mistakes

1. Using the real-world probability instead of p.

2. Discounting by the whole T at each step instead of Δt.

3. For American puts, forgetting to compare with early exercise at every node.

Try before checking

A call has strike $100. Stock can end at $120 or $80. Payoffs?

1. $20 or $0

2. $120 or $80

3. $20 or -$20

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_01_Binomial_trees.pdf; Hull 8e, Chapter 12

Lesson 10: Black-Scholes and the Greeks

Black-Scholes gives a formula for a European option price from five inputs. The Greeks say how the price moves when one input moves.

Start with an everyday idea

Black-Scholes is a pricing recipe for a European option under specific assumptions. The Greeks ask how that recipe’s answer changes when one input changes a little.

Words you need

European means exercise only at expiry. Sigma (σ) is volatility. Delta measures sensitivity to asset price; vega to volatility. The model’s assumptions can fail in real markets.

Helpful earlier topics: Binomial trees for options

Build the explanation

Idea behind it: hold the option and short Δ shares. Over a tiny instant the randomness cancels, so the hedge must earn the risk-free rate. That gives a PDE whose answer is the formula. Prof. Shimko's "Simpler Option Pricing" lecture reaches the same formula with only college calculus.

Inputs: stock S, strike K, rate r, volatility σ, time T. Note: the expected return μ is not an input.

Delta N(d1): change in option price per $1 move in the stock. Also the hedge ratio.

Gamma: how fast delta changes. High near the strike close to expiry.

Theta: value lost per day as time passes. Vega: change per 1% more volatility. Rho: change per 1% higher rate.

Put-call parity: c + Ke-rT = p + S. Use it to check your answer.

Formula and units

$$c = S \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$$

$$p = K \cdot e^{(-rT)} \cdot N(-d2) - S \cdot N(-d1)$$

$$d1 = [\ln(S/K) + (r + \sigma^2/2) \cdot T] / (\sigma\sqrt{T})$$

$$d2 = d1 - \sigma\sqrt{T}$$

$$\text{Delta(call)} = N(d1) \quad \text{Gamma} = \varphi(d1)/(S \cdot \sigma \cdot \sqrt{T}) \quad \text{Vega} = S \cdot \varphi(d1) \cdot \sqrt{T}$$

N = cumulative normal (NORM.S.DIST(x,TRUE) in Excel), φ = normal density.

Walk through the sketch

1. At expiry a call with strike $40 is worth max(S - 40, 0): a hockey stick.

2. Before expiry the price is a smooth curve above the stick. The gap is time value.

3. At S = $42 with 6 months left, Black-Scholes says $4.76.

4. The slope of the curve there is delta = 0.78. Move the stock $1, the call moves about $0.78.

5. With 3 months left (dashed) the curve sits lower and hugs the stick. That loss over time is theta. How sharply the curve bends is gamma.

Worked example

S = 42, K = 40, r = 10%, σ = 20%, T = 0.5 years (Hull's example). Find the call, the put and delta.

1. $\sigma\sqrt{T} = 0.2 \times 0.7071 = 0.1414$

2. $d1 = [\ln(1.05) + (0.10 + 0.02) \times 0.5] / 0.1414 = (0.0488 + 0.06)/0.1414 = 0.7693$

3. $d2 = 0.7693 - 0.1414 = 0.6278$; $N(d1) = 0.7791$, $N(d2) = 0.7349$

4. $c = 42 \times 0.7791 - 40 \times e^{(-0.05)} \times 0.7349 = 4.76$

5. Put by parity: $p = 4.76 - 42 + 40e^{(-0.05)} = 0.81$. Delta = 0.779

Call $4.76, put $0.81, delta 0.78

Common mistakes

1. Using σ^2 where σ goes, or forgetting $\sqrt{T}$.

2. Using a non-continuous rate in $e^{(-rT)}$.

3. Thinking the stock's expected return changes the price. It does not.

Try before checking

Delta is 0.6. For a small $1 stock-price rise, the option changes approximately…

1. $0.60

2. $1.60

3. $6

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_12_Simpler_Option_Pricing_Update_2026.pdf; FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf; Hull 8e, Chapters 14 and 18

Paper lab: Move money through time

A promised $121 arrives in two years. The annual rate is 10%. Move it backward one year at a time.



Predict and try

1. Predict: should today’s value be greater or smaller than $121?

2. Compute PV for rates 0%, 5% and 10%. Keep the two-year date fixed.

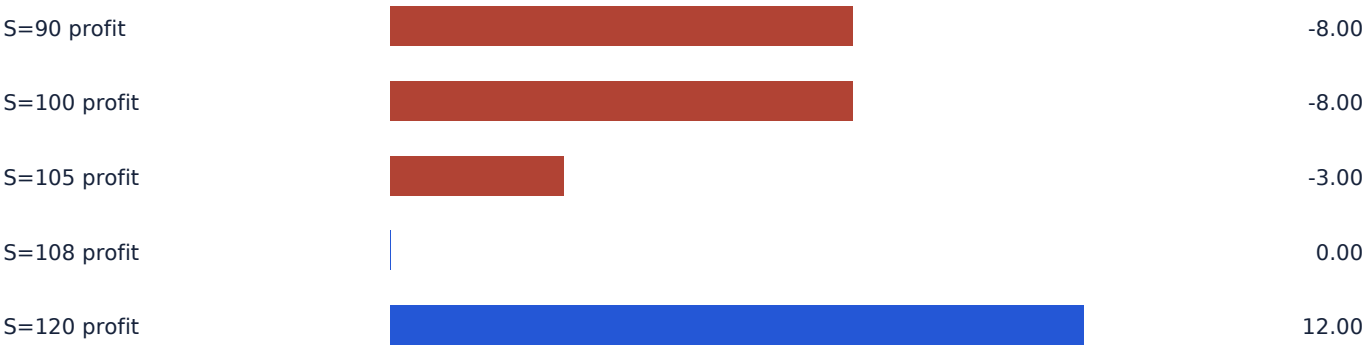
Compare with the explanation

PV = 121/(1+r)^2: $121.00 at 0%, $109.75 at 5%, $100.00 at 10%. A higher discount rate lowers PV. Check: $100 × 1.1 × 1.1 = $121.

Model assumptions: Known future payment; annual compounding; same rate for both years.

Paper lab: Keep option payoff and profit separate

Buy a call on one share with strike $100 and premium $8. Ignore financing of the premium.



Predict and try

1. At a final stock price of $105, would you exercise? Would the whole trade make money?

2. Change the premium to $12. Find the new break-even stock price.

Compare with the explanation

At $105 exercise gives a $5 payoff, but profit = $5 - $8 = -$3. Exercising is better than letting the call expire; it does not recover the premium. Break-even is strike + premium: $108 initially and $112 after the change.

Model assumptions: European call held to expiry; one share; no fees, taxes or interest on the premium.

Paper lab: Rebuild a forward from stock and debt

Buy a $100 share using a loan at 5% continuously compounded. It pays $2 in six months. Deliver the share after one year.



Predict and try

1. Predict: does a dividend raise or lower the fair delivery price?

2. Grow the loan to six months, subtract the dividend, then grow the remaining debt to one year.

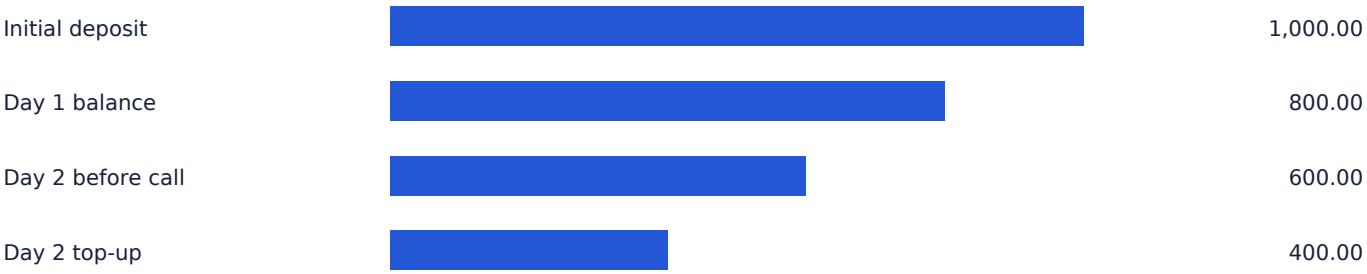
Compare with the explanation

Debt after dividend = 100 exp(0.025) - 2 = $100.5315. At delivery it is 100.5315 exp(0.025) = $103.0765. Without the dividend it would be $105.1271. The dividend lowers the debt you must recover.

Model assumptions: Known dividend; equal borrowing/lending rates; frictionless replication. A forward delivery price is not the current value of an existing forward.

Paper lab: Keep a daily futures ledger

Go long 100 units at quote 100. Deposit $1,000 initial margin. Maintenance is $750. Settlement quotes are 98 and 96.



Predict and try

1. Compute daily trading P/L, then check the maintenance threshold.

2. After restoring initial margin, did the $400 trading loss disappear? Repeat for a short position.

Compare with the explanation

Long: daily P/L is -$200 then -$200. Day 1 balance $800 needs no top-up. Day 2 balance $600 triggers a $400 deposit to restore $1,000. Total deposits $1,400 plus cumulative loss -$400 = balance $1,000. Short: daily gains $200 and $200, final balance $1,400, no calls.

Model assumptions: No interest, fees or withdrawals. A call occurs strictly below maintenance and restores initial margin.

Lesson-check answers

1. The four derivatives and their payoffs

Answer: $20. You may buy for $100 something worth $120: a $20 payoff. Profit also subtracts what you paid for the option.

2. How often interest is added

Answer: $121. Year 1: 100 × 1.10 = 110. Year 2: 110 × 1.10 = 121.

3. Forward price by no-arbitrage

Answer: $105. Borrow $100, buy the asset, and repay $105 in a year. That replication gives a $105 delivery price.

4. Futures, margin and daily settlement

Answer: $20 loss. A long position loses when price falls. (98 - 100) × 10 = -$20.

5. Hedging with futures

Answer: Sell futures. A short futures position tends to gain as futures prices fall, offsetting losses on the asset. Basis risk may remain.

6. Forward rates, duration and convexity

Answer: -5%. Approximate proportional change = -5 × 0.01 = -0.05, or -5%. This is an estimate, not exact repricing.

7. Swaps: trading payment streams

Answer: $10 received. Receive $60, pay $50: net receive $10. The notional itself is not the net payment.

8. Option strategies and put-call parity

Answer: $7. Profit = payoff - premium = 12 - 5 = $7. Payoff alone ignores the purchase cost.

9. Binomial trees for options

Answer: $20 or $0. A call can be left unused. Its payoff is max(stock price - strike, 0).

10. Black-Scholes and the Greeks

Answer: $0.60. Delta × stock change = 0.6 × $1 = $0.60, holding other inputs fixed. Delta itself can change.

Derivatives midterm-style practice

5 questions | 40 points | Learn at your own pace.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Forward price with growing quarterly dividends (8 points)

2. Recover spot from a futures loss (8 points)

3. Find the no-margin-call boundary (8 points)

4. Value an existing bond forward (8 points)

5. Find the fair fixed rate on a swap (8 points)

Question 1: Forward price with growing quarterly dividends

8 points | October 17 paper Q1 · Equity forwards and no-arbitrage

From your paper: stock price $200, continuous risk-free rate 4%, delivery in 3 years. Quarterly dividends begin in 3 months at $1.50; each is 1% above the previous one. Find the fair forward delivery price. State how you treat the dividend at the delivery date.

Show your setup, units, calculation and interpretation.

Question 2: Recover spot from a futures loss

8 points | October 17 paper Q2 · Index futures and dividend yield

From your paper: a long one-year index futures contract has notional 200 × F. Initial index 1,100, continuous dividend yield 4%, risk-free rate 2%. Three months later the investor has lost $100, with no interest on the margin account. Find the index level then.

Show your setup, units, calculation and interpretation.

Question 3: Find the no-margin-call boundary

8 points | October 17 paper Q3 · Daily settlement and collateral

From your paper: short 20 futures contracts, each for 250 index units, initial delivery quote 1,500. Initial margin is 5% of notional; maintenance is 90% of initial margin. Margin earns 4% continuously. Day 1 settlement is 1,498; day 2 is X. Find the largest X with no margin top-up on day 2. State a day-count and interest-timing assumption.

Show your setup, units, calculation and interpretation.

Question 4: Value an existing bond forward

8 points | October 17 paper Q4 · Known income and rate conventions

From your paper: a 10-year bond is worth $920 today. You hold a long forward delivering in one year at $940. The bond pays $80 every six months, including a payment in six months and another just before delivery. Six-month and one-year spot rates are respectively 7% and 8%, quoted annually with semiannual compounding. Find the current contract value.

Show your setup, units, calculation and interpretation.

Question 5: Find the fair fixed rate on a swap

8 points | October 17 paper Q5 · Term structure and swaps

From your paper: annual-compounded spot rates for years 1-6 are 7.0%, 7.3%, 7.7%, 8.1%, 8.4% and 8.8%. A six-year, $10 million plain-vanilla swap pays annually; each floating rate is fixed at the start of its year. Find the fixed annual rate that makes the two legs equal.

Show your setup, units, calculation and interpretation.

Solution 1: Forward price with growing quarterly dividends

First, the words

A forward price is the delivery price that gives a new contract zero value. Known dividends belong to the stock holder before delivery, so remove their present value before growing the stock’s remaining value to delivery.

A smaller problem first

If a stock costs $100 and pays a known $2 dividend before delivery, the forward buyer will not receive that dividend. Subtract its value today first. With continuous interest, moving money forward uses exp(rt); moving it backward uses exp(-rt). A quarter is 0.25 years.

Hint

List twelve dates: 0.25, 0.50, …, 3.00 years. Dividend k is 1.50 × 1.01^(k-1). Discount each dividend separately; then apply the forward-price formula.

Work through it

1. Assume all 12 dividends, including the one at year 3, are paid before delivery. If delivery precedes the final dividend, exclude that payment and explain the different convention.

2. For k = 1,...,12, set tk = k/4 and Dk = 1.50 × 1.01^(k-1). The first dividend is $1.50, not $1.515.

3. Dividend present value I = Σ Dk exp(-0.04tk) = $17.81603. This is a finite sum; the dividends grow but their dates still need separate discounting.

4. Forward price F0 = (S0 - I) exp(rT) = (200 - 17.81603) exp(0.04 × 3) = $205.41185, about $205.41.

5. Check: without dividends the forward would be $225.50. Dividends lower the delivery price. If the year-3 dividend is excluded, add its terminal amount, approximately $1.67, to the result.

6. Transfer drill: if the dividend pattern stays fixed but S0 rises $10, F0 rises 10 × exp(0.12), about $11.27. Explain this without recalculating all dividends.

Mark your reasoning

1. 2: dates, dividend growth and timing assumption

2. 2: discounted dividend sum

3. 3: no-arbitrage formula and $205.41

4. 1: economic direction check

Syllabus: equity forwards, arbitrage and fair-value derivation. Hull 8e Chapter 5: known-income assets. Source photo page 1. Solution is independently derived.

My mark: ____ / 8. My next correction: _____

Solution 2: Recover spot from a futures loss

First, the words

A contract multiplier converts a one-point futures move into a dollar gain or loss. Spot is the index now; futures is its delivery-price quote. Dividend yield q is a continuous proportional payout rate, unlike a known dollar dividend.

A smaller problem first

For a long contract with multiplier 200, a 0.5-point futures fall loses $100: 200 × (-0.5). After three months, a one-year contract has nine months, or 0.75 years, remaining.

Hint

Find the original futures quote using F0 = S0 exp((r-q)T). Use the loss and multiplier to find the new futures quote. Only then convert it back to spot using remaining maturity.

Work through it

1. Assume one contract, multiplier 200 dollars per index point, unchanged rates/yield and the usual deterministic-rate spot/futures relation.

2. Original futures quote F0 = 1,100 exp((0.02 - 0.04) × 1) = 1,078.21854. A yield greater than financing cost puts this quote below spot.

3. Cumulative settlement loss: 200(Ft - F0) = -100. Therefore Ft = F0 - 0.5 = 1,077.71854.

4. Remaining maturity is 1 - 0.25 = 0.75 years. Ft = St exp((0.02 - 0.04) × 0.75).

5. St = 1,077.71854 exp(0.015) = 1,094.00617, about 1,094.01. Do not subtract 100 directly from the index.

6. Transfer drill: for a short position with a $100 loss, the futures quote instead rises 0.5 points. Explain both the sign change and the remaining-time conversion.

Mark your reasoning

1. 2: original quote

2. 2: multiplier and sign of loss

3. 2: remaining maturity and inversion

4. 2: 1,094.01 with stated assumptions

Syllabus: futures mechanics, uses and equity fair-value pricing. Hull 8e Chapters 2 and 5. Source photo page 1. Margin interest is explicitly zero in this question.

My mark: ____ / 8. My next correction: _____

Solution 3: Find the no-margin-call boundary

First, the words

Initial margin is the opening collateral. Maintenance margin is the threshold below which more cash must be added. A short futures position gains when the settlement quote falls. Interest earned on collateral is a separate cash flow.

A smaller problem first

If you are short 2 contracts, each worth $10 per point, a 3-point fall gains 2 × 10 × 3 = $60. A later rise reverses part of that gain. Keep a running balance rather than treating each day in isolation.

Hint

The total multiplier is 20 × 250 = 5,000. Begin with margin, add a day of interest and day 1’s short-position gain. Carry that balance to day 2 and solve an inequality against maintenance margin.

Work through it

1. Assume one day is 1/365 year, interest accrues on the previous end-of-day balance before each daily settlement, and the stated index values are the settlement quotes used for marking to market. The paper does not specify the day-count convention.

2. Notional = 20 × 250 × 1,500 = $7,500,000. Initial margin M0 = 5% × 7,500,000 = $375,000. Maintenance = 90% × 375,000 = $337,500.

3. Day 1 gain for the short = 5,000 × (1,500 - 1,498) = $10,000. M1 = 375,000 exp(0.04/365) + 10,000 = $385,041.10.

4. Day 2 balance M2 = M1 exp(0.04/365) + 5,000(1,498 - X). No call requires M2 ≥ 337,500.

5. Solve at equality: Xmax = 1,498 + [M1 exp(0.04/365) - 337,500]/5,000 = 1,507.51666, about 1,507.52.

6. Check: an increase above Xmax lowers a short’s margin balance below the threshold. If quotes must be on a specified tick grid, round downward to the largest allowed quote.

7. Transfer drill: without margin interest, the boundary is 1,507.50. With a different stated day fraction δ, replace both 1/365 terms by δ. Do not hide the convention.

Mark your reasoning

1. 2: initial and maintenance margin

2. 2: daily short-position gains/losses

3. 2: interest-aware inequality

4. 2: boundary and explicit conventions

Syllabus: futures market mechanics and fair-value applications. Hull 8e Chapter 2; supplied margin lecture. Source photo page 1. Different day-count assumptions produce slightly different numerical boundaries.

My mark: ____ / 8. My next correction: _____

Solution 4: Value an existing bond forward

First, the words

The old delivery price K stays fixed even when the fair price for a new contract changes. Contract value is what the existing position is worth today. A nominal annual rate compounded semiannually is divided by 2 for each half-year period.

A smaller problem first

If the fair new delivery price is $95 but your long contract forces you to pay $100, your contract is worth a negative amount. Discount the $5 disadvantage back to today. Coupon income before delivery belongs to the bond holder, not the forward buyer.

Hint

Subtract the present values of the two coupons from the bond price, then subtract the discounted delivery payment. Use the six-month rate for the first coupon and the one-year rate for the year-one cash flows.

Work through it

1. Assume “own a forward” means a long position and $920 is the cash/dirty bond price used for valuation. If it were a clean quote, accrued interest would be required; it is not provided.

2. Six-month discount factor = $1/(1 + 0.07/2) = 1/1.035$. First coupon PV = $80/1.035 = \$77.29469$.

3. One-year discount factor = $1/(1 + 0.08/2)^2 = 1/1.04^2$. Second coupon PV = $73.96450; delivery-payment PV = $940/1.04^2 = \$869.08284$.

4. Long forward value f0 = 920 - 77.29469 - 73.96450 - 869.08284 = -$100.34202, about -$100.34.

5. Cross-check: the fair new forward price is $(920 - 77.29469 - 73.96450) \times 1.04^2 = \$831.47007$. Then f0 = $(831.47007 - 940)/1.04^2$.

6. Transfer drill: if K falls $10 with everything else unchanged, the long’s value rises $10/1.04^2$, about $9.25. The bond’s 10-year maturity does not make the forward a 10-year contract.

Mark your reasoning

1. 2: two correct discount factors

2. 2: remove both pre-delivery coupons

3. 3: current long value -$100.34

4. 1: distinguish contract value from new forward price

Syllabus: equity/interest-rate forwards, rate conventions and arbitrage pricing. Hull 8e Chapters 4-5. Source photo page 1. The final coupon is explicitly before delivery.

My mark: ____ / 8. My next correction: _____

Solution 5: Find the fair fixed rate on a swap

First, the words

A par swap starts with equal fixed-leg and floating-leg values. Discount factors value $1 at different dates. The par rate is a weighted rate across the payment dates, not the last spot rate or an arithmetic average.

A smaller problem first

For a one-year swap with one payment and discount factor D1, K = (1 - D1)/D1. If D1 = 1/1.05, K is 5%. With several payment dates, add their discount factors in the denominator.

Hint

For each year t calculate Dt = (1 + st)^(-t). With annual accrual and the standard single-curve par-swap assumptions, K ΣDt = 1 - D6.

Work through it

1. Assume a new swap at a reset date, annual accrual factors equal to 1, a common curve for discounting and forwarding, and no credit/funding adjustment.

2. Discount factors D1…D6 are approximately 0.934579, 0.868561, 0.800484, 0.732314, 0.668119 and 0.602874.

3. Under these assumptions, floating coupon-leg PV per dollar of notional is 1 - D6. Fixed coupon-leg PV is K(D1 + … + D6).

4. Equate the legs: K = (1 - 0.6028741017)/(0.9345794393 + 0.8685614624 + 0.8004844320 + 0.7323138101 + 0.6681186409 + 0.6028741017).

5. K = 0.08620181676, or about 8.62018% annually. Fixed yearly cash payment is approximately $862,018.17 on $10 million.

6. Check: notional cancels from the fair-rate equation. Doubling notional doubles payments but leaves K unchanged.

7. Transfer drill: with semiannual payments, use half-year accrual factors and a curve at those payment dates. Do not simply halve the annual par rate without repricing.

Mark your reasoning

1. 2: rate convention and discount factors

2. 2: correct fixed/floating PV equation

3. 3: 8.62018% and payment interpretation

4. 1: assumptions and notional cancellation

Syllabus: swaps, applications and theoretical fair value. Hull 8e Chapters 4 and 7. Source photo page 2. This is the classroom single-curve framework.

My mark: ____ / 8. My next correction: _____

Derivatives mock final

8 questions | 100 points | 120 minutes suggested practice time.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Price a forward and test an arbitrage (12 points)
2. Separate margin deposits from trading profit (12 points)
3. Set and revalue a plain interest-rate swap (12 points)
4. Use parity and distinguish payoff from profit (12 points)
5. Work backward through an American option tree (12 points)
6. Connect a delta hedge to Black-Scholes (14 points)
7. Scale Brownian shocks and interpret log returns (14 points)
8. Explain what a delta hedge leaves behind (12 points)

Question 1: Price a forward and test an arbitrage

12 points | Syllabus: forward fair value and arbitrage; extends midterm Q1.

A stock is $100. It pays a known $2 dividend in six months. The continuously compounded risk-free rate is 5% and delivery is in one year. Find the fair forward price. A dealer instead quotes $106. Describe the cash-and-carry trade and its terminal profit per share. Assume borrowing/lending at 5%, no fees, and the dividend is certain.

Show your setup, units, calculation and interpretation.

Question 2: Separate margin deposits from trading profit

12 points | Syllabus: futures mechanics; extends midterm Q2-Q3.

You short two futures contracts at 4,000. Each has a $50 multiplier per index point. Total initial margin is $2,000 and total maintenance margin is $1,500. Settlement quotes are 4,003 on day 1 and 4,006 on day 2. Margin earns no interest. Show daily P/L, balances before deposits, any required deposit, cumulative P/L and total deposits. Calls occur strictly below maintenance.

Show your setup, units, calculation and interpretation.

Question 3: Set and revalue a plain interest-rate swap

12 points | Syllabus: swap applications and fair value; extends midterm Q5.

A three-year swap has annual payments, $1m notional and discount factors D1 = 0.96, D2 = 0.91, D3 = 0.85. Calculate the par fixed rate K. In a hypothetical immediate repricing before any floating coupon has fixed, the factors become 0.95, 0.88, 0.80. Find the value to the fixed-rate payer. Assume the floating leg resets at valuation, a single projection/discount curve, and unit accrual periods.

Show your setup, units, calculation and interpretation.

Question 4: Use parity and distinguish payoff from profit

12 points | Syllabus: option basics and strategies.

A non-dividend-paying stock is $100. A European call with strike $100 costs $9 and the strike discount factor is 0.95. Find the matching put price and explain put-call parity as two equal terminal portfolios. Separately, a bull call spread buys a strike-90 call for $15 and sells a strike-110 call for $4. Find its payoff and simple profit at stock prices 80, 100 and 120, plus maximum loss and gain. Ignore financing of premiums for the spread.

Show your setup, units, calculation and interpretation.

Question 5: Work backward through an American option tree

12 points | Syllabus: option modeling and fair value; extension beyond the midterm.

A stock starts at $100. Each of two periods multiplies it by u = 1.2 or d = 0.8. The risk-free growth factor is R = 1.05 each period. Price a strike-$105 American put and the corresponding European put. Show the terminal payoffs, pricing probability, continuation values and early-exercise decision. Assume no dividends and a frictionless tree.

Show your setup, units, calculation and interpretation.

Question 6: Connect a delta hedge to Black-Scholes

14 points | Syllabus: fair-value derivation, Black-Scholes and Greeks.

Assume $dS = \mu S\,dt + \sigma S\,dW$ and a smooth European option $V(S,t)$, no dividends, frictionless continuous hedging and constant r, σ. Using Ito's lemma, explain why a position long V and short V_S shares earns r locally and derive the BSM PDE. Then value a one-year call with $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, given $N(d1) = 0.63683$, $N(d2) = 0.55962$ and $\exp(-0.05) = 0.951229$. State delta and two model limitations.

Show your setup, units, calculation and interpretation.

Question 7: Scale Brownian shocks and interpret log returns

14 points | Syllabus: Wiener processes underlying option models.

For geometric Brownian motion with S0 = 100, μ = 6% annually and σ = 20% annually, use Ito to give the distribution of ln(S_T/S0). At T = 0.25 years, find its mean and standard deviation. Calculate S_T for a standard-normal draw Z = -1. Explain which drift changes under risk-neutral valuation if r = 4% and there are no dividends.

Show your setup, units, calculation and interpretation.

Question 8: Explain what a delta hedge leaves behind

12 points | Syllabus: option Greeks and derivative uses.

A dealer is short 100 calls, each covering 100 shares. Each per-share call has delta 0.60, gamma 0.02 per dollar, and vega $0.15 per one percentage-point volatility increase. How many shares make the position initially delta neutral? Estimate the hedged P/L for a $2 stock rise at unchanged volatility using $0.5\,\Gamma(\Delta S)^2$. Separately estimate P/L for a 3-percentage-point volatility rise at unchanged stock price. Ignore time decay, funding and higher-order terms. Explain the hedge limitations.

Show your setup, units, calculation and interpretation.

Solution 1: Price a forward and test an arbitrage

First, the words

A forward fixes a future purchase price. A known dividend belongs to the stock owner before delivery. Arbitrage means matching cash flows with no net risk under the stated assumptions.

A smaller problem first

Without dividends, buying a $100 stock with a 5% continuously compounded loan creates a one-year cost of 100 exp(0.05) = $105.13.

Hint

Borrow the stock price today; use the dividend to reduce the loan at six months.

Work through it

1. Dividend PV = 2 exp(-0.05 × 0.5) = $1.9506.

2. Fair delivery price = (100 - 1.9506) exp(0.05) = $103.0765.

3. Borrow $100, buy one share, and sell a one-year forward at $106. Net initial cash is zero.

4. At six months use the $2 dividend to reduce the debt. At maturity deliver the share for $106 and repay 100 exp(0.05) - 2 exp(0.025) = $103.0765.

5. Locked terminal profit = $2.9235. This relies on identical timing and frictionless funding.

Mark your reasoning

1. 3: dividend timing and PV

2. 3: fair forward price

3. 4: all trade cash flows

4. 2: profit and assumptions

Hull: forward prices with known income; supplied forward-pricing lecture.

My mark: ____ / 12. My next correction: _____

Solution 2: Separate margin deposits from trading profit

First, the words

Short means you gain when the futures quote falls. Maintenance margin is the trigger; initial margin is the target after a call.

A smaller problem first

A short position of 100 units loses $100 when the quote rises by $1. A deposit of $100 covers collateral but does not undo that loss.

Hint

Daily P/L = -2 × 50 × (new quote - previous quote).

Work through it

1. Total exposure is 100 dollars per point; use the short-position minus sign.

2. Day 1: P/L = -$300; balance = $1,700. No call.

3. Day 2: P/L = -$300; balance before deposit = $1,400, below $1,500.

4. Deposit $600 to restore $2,000. Total deposits, including the original $2,000, are $2,600.

5. Cumulative trading P/L = -$600. Check: deposits $2,600 plus P/L -$600 = final balance $2,000.

Mark your reasoning

1. 2: exposure and sign

2. 3: day 1

3. 3: day 2 and margin call

4. 4: cumulative loss and balance identity

Hull: futures markets and margin; supplied futures-markets lecture.

My mark: ____ / 12. My next correction: _____

Solution 3: Set and revalue a plain interest-rate swap

First, the words

A discount factor is the price today of $1 paid later. A par swap has zero value at inception. A payer swap pays fixed and receives floating.

A smaller problem first

If annual discount factors are 0.96 and 0.90, a two-year par fixed rate is (1 - 0.90)/(0.96 + 0.90).

Hint

At reset, PV of floating coupons is N(1 - D3); PV of fixed coupons is NK times the sum of discount factors.

Work through it

1. Original annuity factor = 0.96 + 0.91 + 0.85 = 2.72.

2. K = (1 - 0.85)/2.72 = 0.05514706, or 5.5147% annually.

3. After repricing, floating coupon PV = 1,000,000 × (1 - 0.80) = $200,000.

4. Fixed coupon PV = 1,000,000 × K × 2.63 = $145,036.76.

5. Payer value = floating minus fixed = +$54,963.24. Away from a reset, account for the already-fixed next floating coupon.

Mark your reasoning

1. 3: original annuity and par rate

2. 3: floating PV

3. 3: fixed PV

4. 3: payer sign and reset assumption

Hull: interest-rate swaps; supplied duration/swaps lecture.

My mark: ____ / 12. My next correction: _____

Solution 4: Use parity and distinguish payoff from profit

First, the words

Payoff is the contract cash at expiry. Profit subtracts its initial cost; this question explicitly ignores financing of that cost.

A smaller problem first

A call with strike $100 has payoff $20 at a stock price of $120. If its premium was $8, simple profit is $12.

Hint

Parity: call + discounted strike = put + stock. Spread payoff is the difference of two call payoffs.

Work through it

1. Put = 9 + 100 × 0.95 - 100 = $4.

2. Call plus a bond paying $100 and stock plus put both pay max(stock price, 100) at expiry; their prices must match.

3. Spread premium paid = 15 - 4 = $11. Payoff = max(S - 90, 0) - max(S - 110, 0).

4. At S = 80, 100, 120: payoffs are $0, $10, $20; simple profits are -$11, -$1, +$9.

5. Maximum loss is $11; maximum gain is $9. Profit and payoff are different graphs.

Mark your reasoning

1. 3: put price

2. 3: replication explanation

3. 4: three payoffs and profits

4. 2: maximum gain/loss

Hull: properties of stock options and trading strategies.

My mark: ____ / 12. My next correction: _____

Solution 5: Work backward through an American option tree

First, the words

Risk-neutral probabilities price tradable claims; they need not be forecasts. An American option allows early exercise.

A smaller problem first

At any American put node, compare the discounted continuation value with immediate exercise max(K - S, 0). Keep the larger.

Hint

q = (R - d)/(u - d). The period-1 down node is the important exercise decision.

Work through it

1. q = (1.05 - 0.8)/(1.2 - 0.8) = 0.625; 0 < q < 1.

2. Terminal stock prices are 144, 96, 64. Put payoffs are 0, 9, 41.

3. At S = 120, continuation = (0.625 × 0 + 0.375 × 9)/1.05 = $3.2143; immediate exercise is 0.

4. At S = 80, continuation = (0.625 × 9 + 0.375 × 41)/1.05 = $20; immediate exercise is $25. American exercise wins.

5. American root value = (0.625 × 3.2143 + 0.375 × 25)/1.05 = $10.8418, greater than immediate exercise $5.

6. European root uses $20 at the down node: value = $9.0561. Early-exercise flexibility is worth $1.7857.

Mark your reasoning

1. 2: q

2. 2: terminal payoffs

3. 4: both continuation/exercise comparisons

4. 2: American root

5. 2: European comparison

Hull: binomial trees and American exercise; supplied Binomial_trees.pdf. This adds depth beyond the short app lesson.

My mark: ____ / 12. My next correction: _____

Solution 6: Connect a delta hedge to Black-Scholes

First, the words

Delta is the local change in option value per $1 stock move. A locally riskless hedge removes the Brownian shock, not all future risk without rebalancing.

A smaller problem first

If a call delta is 0.6, buying 0.6 shares offsets the immediate stock sensitivity of one short call.

Hint

Ito gives $dV = (V_t + \mu S V_S + 0.5\sigma^2S^2V_{SS})dt + \sigma S V_S dW$. Subtract $V_S dS$.

Work through it

1. The self-financing hedged change is $dV - V_S dS = (V_t + 0.5\sigma^2S^2V_{SS})dt$; the dW and μ terms cancel.

2. The hedge value is $V - S V_S$. No arbitrage requires its drift to equal $r(V - S V_S)$.

3. Rearrange: $V_t + 0.5\sigma^2S^2V_{SS} + rS V_S - rV = 0$, with terminal condition $V(S,T) = \max(S - K, 0)$ for a call.

4. $C = S N(d1) - K \exp(-rT) N(d2) = 100 \times 0.63683 - 100 \times 0.951229 \times 0.55962 \approx \$10.45$.

5. Delta = $N(d1) = 0.63683$. For one short option on one share, buy that many shares initially and rebalance.

6. Constant volatility omits smiles and volatility changes; continuous frictionless hedging omits costs, gaps and discrete rebalancing. These limitations matter for actual hedging.

Mark your reasoning

1. 4: cancellation and self-financing reasoning

2. 3: PDE and terminal condition

3. 3: numerical call

4. 2: delta

5. 2: two limitations

Hull: Wiener processes, Ito and the Black-Scholes model. Derivation is an original scaffold, not a reproduced exam answer.

My mark: ____ / 14. My next correction: _____

Solution 7: Scale Brownian shocks and interpret log returns

First, the words

A Brownian increment over length Δt is sqrt(Δt) times a standard-normal draw. Drift scales with time; uncertainty scales with its square root.

A smaller problem first

Annual volatility 20% becomes 10% over a quarter: 20% × sqrt(0.25). It is not 5%.

Hint

For f(S) = ln S, f_S = 1/S and f_SS = -1/S². Include the second-order Ito term.

Work through it

1. Ito gives d ln S = (μ - 0.5σ²)dt + σ dW.

2. Thus ln(S_T/S0) = (μ - 0.5σ²)T + σ sqrt(T) Z, with Z standard normal.

3. Quarter-year mean = (0.06 - 0.5 × 0.20²) × 0.25 = 0.01. Standard deviation = 0.20 sqrt(0.25) = 0.10.

4. For Z = -1, log return = 0.01 - 0.10 = -0.09, so S_T = 100 exp(-0.09) = $91.39.

5. Under the risk-neutral measure replace μ by r = 0.04. Log-return mean becomes 0.005; volatility remains 0.10 in this model.

6. The physical drift describes the assumed real-world process; the pricing drift enforces no-arbitrage valuation. They answer different questions.

Mark your reasoning

1. 4: Ito log transformation

2. 3: distribution and time scaling

3. 3: numerical price

4. 2: risk-neutral drift

5. 2: interpretation

Hull: Wiener processes and Ito; Valuation stochastic-process lecture for supporting intuition.

My mark: ____ / 14. My next correction: _____

Solution 8: Explain what a delta hedge leaves behind

First, the words

Gamma measures how delta changes as the stock moves. Vega measures sensitivity to a volatility change. Here vega is quoted per one percentage point of volatility.

A smaller problem first

A share has delta 1 and gamma 0. Shares can neutralize delta but cannot cancel option gamma.

Hint

There are 10,000 option-share units. Short positions reverse the Greeks' signs.

Work through it

1. Option position delta = -10,000 × 0.60 = -6,000. Buy 6,000 shares.

2. Total option gamma = -10,000 × 0.02 = -200 per dollar. The stock adds zero gamma.

3. After canceling the first-order delta term, estimated stock-move P/L = 0.5 × (-200) × 2^2 = -$400.

4. Total vega = -10,000 × 0.15 = -$1,500 per volatility percentage point. A 3-point increase gives approximately -$4,500.

5. Delta neutrality is local and temporary. Gamma changes delta, volatility changes option prices, and discrete hedging/costs prevent a perfect continuous hedge.

Mark your reasoning

1. 3: multiplier and share hedge

2. 3: gamma P/L

3. 3: vega units and P/L

4. 3: limitations

Hull: Greeks and hedging. Supplied syllabus lists Greeks in the final option block.

My mark: ____ / 12. My next correction: _____